

Highlights

Two bases for a voxel, the medium and the wavefield: the error of elastic volume-integral scattering, isolated on a space-filling layer of cubes

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- A voxel needs two bases: one for the medium and one for the wavefield
- On a layer of cubes the discretisation error is isolated, with no shape error
- The order is $\min(2p+2, 2r+2)$ for field degree p and medium degree r
- The single-site T-matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

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Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral (coupled-dipole) methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves, which cannot be measured where no exact solution exists. We answer on a layer of voxels: because cubes tile space, n planes of voxels are the layer itself, with no shape error, and its exact solution isolates the discretisation error. The source-averaged coupling equals the continuum's exactly; the cube's single-site T -matrix follows from distributional moments of the Green's tensor. With the wavefield expanded to Legendre degree p and the medium's contrast to degree r in each voxel, the scheme converges at order $\min(2p+2, 2r+2)$, with leading error $c_p(kd)^{2p+2}$ in closed form: $c_0 = 1/12$, $c_1 = 1/720$, $c_2 = 1/100800$; for $p \leq 1$ also for SV and SH waves and at oblique incidence. The single-site T -matrix's second-order error equals the relative projection error of the medium at degree $\min(p, r)$, exactly in the long-wave limit: the two bases must be matched, and the error

is known from the medium before anything is solved. Where the medium is piecewise constant on the voxels it vanishes: in a randomly stratified layer one first-moment voxel per layer reaches 10^{-6} . A smoothly graded sphere converges at second order against its exact solution. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background, which the model prescribes. The other is the *wavefield*: the displacement and strain

inside the voxel, which the calculation must find. Each is expanded in a basis on the cell, and the two bases are separate choices. Here both are Legendre polynomials, the wavefield to degree p and the contrast to degree r ; the usual scheme, a uniform field in a cell of constant properties, is $p = r = 0$. This paper asks which basis each of the two needs, and what error a given pair (p, r) leaves. The answer decides the cost of everything built on it, because a voxel that represents its contents more accurately can be larger, and the number of voxels, not the size of each one's state, sets the cost of the multiple-scattering solve.

Why the question is hard to answer. For the media such a scheme is built for, there is no exact solution. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green's tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the single voxel's response is matched to cancel it, refinement converges to the wrong answer without revealing it.

The method: a layer of voxels. This paper therefore answers the question in the simplest configuration in which a voxel scheme runs in full (every voxel, every coupling, the whole multiple-scattering series, at normal and at oblique incidence) and the exact answer is nevertheless known: a layer filled with voxels. Because cubes tile space, a layer of identical voxels does not approximate a homogeneous layer; it *is* one. Any difference between its reflection and transmission and those of the continuous layer is therefore

42 error of the discretisation alone, with no modelling error and no shape error
 43 behind which to hide. A sphere, the usual test object for volume-integral and
 44 discrete-dipole methods [1], cannot give this: its error mixes the discretisation
 45 with the staircase approximation of a curved surface by cubes. The layer
 46 plays the part of the patch test in finite elements: reproducing a uniform
 47 medium correctly is necessary, though not sufficient, for a scheme to be
 48 trusted in a heterogeneous one. The experiment is then made harder in
 49 three steps, each with an exact solution of its own: a randomly stratified
 50 layer, in which every cell is homogeneous but the cells differ; a layer whose
 51 properties vary smoothly within every cell, which is where the basis for the
 52 medium is tested; and a sphere whose contrast falls smoothly to zero at its
 53 surface, which carries the layer's prediction into three dimensions without a
 54 staircase. What the layer establishes about the scheme's two ingredients, the
 55 coupling between voxels and the response of a single voxel, are properties of
 56 the ingredients, which are the same in any medium. A laterally heterogeneous
 57 medium remains to be tested.

58 *The answer.* (1) *The order is set by the weaker basis.* The scheme converges
 59 at order $\min(2p + 2, 2r + 2)$ in the cell size (§9). A contrast held constant in
 60 each cell therefore caps every voxel at second order in a medium that varies
 61 within the cells, however rich the basis of its field; and a contrast of higher
 62 degree than the field is wasted. (2) *The leading error is in closed form, for*
 63 *every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$
 64 with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§8); the first-moment voxel,
 65 $p = 1$, keeps its fourth order and its constant for P, SV and SH waves at
 66 normal and oblique incidence. For the collocation scheme, the uniform-field

67 voxel in its simplest form, the error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$
 68 in transmission (§7), and the opposite signs explain why no correction of the
 69 single site can improve both (§11). (3) *The error that remains is the single*
 70 *site's, and it is the projection error of the medium.* The single-site T -matrix
 71 sums the rescattering within a voxel to all orders, and the coupling between
 72 voxels is exact, so the multiple scattering itself is not approximated. But
 73 the T -matrix is that of the voxel as represented: its contrast projected to
 74 degree r and its field held to degree p . When the scattered field is separated
 75 by its order in the contrast, the error of the second-order term, in which
 76 the field scattered once within a voxel is scattered there again, equals the
 77 *relative projection error* of the medium: the fraction of the squared norm of
 78 its profile that polynomials of degree $\min(p, r)$ on the cells cannot represent
 79 (Eq. (38)). The equality is exact in the long-wave limit, where it is derived,
 80 and holds to within 1.3% for all nine pairs with $p, r \leq 2$ in the experiments
 81 (§9). It shows what the wavefield basis is for: it must carry what the medium
 82 imprints on the field, not only the wave, which at these frequencies varies
 83 little across a cell. (4) *The two bases must be matched, and the error is*
 84 *known in advance.* The relative projection error is a property of the medium
 85 and the grid alone, so the error of a calculation is known before anything is
 86 solved, and the cell size and the degree that a required accuracy needs can
 87 be chosen beforehand. (5) *Where the medium is piecewise constant on the*
 88 *cells the projection error vanishes,* a constant suffices for the medium, and
 89 the degree of the field alone sets the order: in a randomly stratified model
 90 one first-moment voxel per model layer reaches a reflection accuracy of 10^{-6} ,
 91 where the uniform-field voxel needs about forty (§9), and in the uniform

92 layer it allows voxels 58 times larger in each direction (§11). (6) *In three*
 93 *dimensions* the smoothly graded sphere converges at second order with the
 94 mean-only voxel, as the layer predicts (§10), and at fourth order with the
 95 first-moment voxel in the companion paper [2].

96 *What makes the experiment exact.* The answer above can be trusted only if
 97 nothing but the two bases is approximated. Three results, set against the
 98 coupled-dipole method described below, secure that. (a) *An exact coupling.*
 99 The integrated Green tensor and the filtered coupled dipoles approximate the
 100 field of a cell; for cubes the source-cell average is exact, because the cube's
 101 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and
 102 at normal incidence the lattice kernel is in closed form with no lattice sum.
 103 (b) *No polarisability to choose.* The polarisability, a modelling choice in the
 104 coupled-dipole method, is here a derived object (§6): the moments of the
 105 Green's tensor over the cube, taken as distributions, yield Eshelby's δ term
 106 without adding it by hand, the cube's departure from the sphere as a pure
 107 cubic anisotropy, and the closure's error against exact scattering by a sphere
 108 as a real $(ka)^2$ term in closed form. The self term the single site carries is
 109 the one the averaged kernel omits, so the discrete system is a collocation of
 110 the continuum equation. (c) *The elastic state.* A voxel carries a force and
 111 a moment, so its state has nine components, the coupling is a 9×9 tensor,
 112 and its local term is the depolarisation of a plate.

113 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
 114 *space*, so a uniform region discretised into cubes is still that uniform region,
 115 whereas spheres or point scatterers can only approximate it and carry a pack-
 116 ing error of their own. Their *regular lattice* makes the interaction between

117 cells translation invariant, so the coupling is a convolution (Toeplitz between
118 planes, Bloch-periodic within them), and the whole system is solved by fast
119 transforms. And a cube small compared with the wavelength has a *single-site*
120 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
121 approximation a voxel introduces is the internal field it assumes.

122 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of
123 small scatterers that interact through the background Green’s function is the
124 multiple-scattering description of Foldy [3] and Lax [4]. In electromagnetics
125 it became a numerical method as the coupled-dipole method, or discrete
126 dipole approximation (DDA): each cell of a cubic grid is a polarisable point
127 dipole driven by the incident field and by the fields of every other dipole
128 [5, 6]. The method is a collocation of the volume integral equation for the
129 electric field, with the cell’s self-interaction absorbed into its polarisability
130 [7], and Yurkin and Hoekstra [8] review it in that form. Its accuracy turns on
131 two ingredients, and this paper takes up both for elasticity. The first is the
132 polarisability. The Clausius–Mossotti value is the static response of a sphere;
133 the lattice dispersion relation of Draine and Goodman [9] adjusts it so that
134 an infinite lattice propagates waves as the continuum does to a given order
135 in kd ; Rahmani et al. [10] instead derive it from the exact static solution for
136 the whole body, so that the method is exact in the long-wavelength limit.
137 The second is the coupling. Integrating the Green tensor over the source
138 cell rather than sampling it at the centre improves the accuracy, most of
139 all for strong contrasts [11], and the filtered coupled dipoles of Piller and
140 Martin [12], revived by Yurkin et al. [13] for large refractive indices, replace
141 the point dipole by a source of finite extent whose spectrum is band-limited

142 to the grid. The coupling summed over a lattice extends the method to
 143 periodic structures [14]. Its error converges at second order in the dipole
 144 spacing when shape errors are absent [1], and for a cube, the one shape
 145 a cubic grid represents without a staircase, Yurkin and Kahnert [15] reach
 146 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. The
 147 layer studied here is the elastic counterpart of that cube: a body the grid
 148 represents exactly, so that the discretisation error can be isolated. What this
 149 paper adds is the closed form: for cells that tile space, the coupling that
 150 makes the lattice reproduce the continuum exactly, and the order and the
 151 constant of the error that remains, as functions of the two bases. In the
 152 terms used here the coupled-dipole method is the pair $p = r = 0$, and its
 153 refinements improve the polarisability and the coupling while leaving both
 154 bases constant in the cell.

155 *Its elastic counterparts.* The elastic problem has the same volume integral
 156 equation [16], and every cell-based solver of it is a coupled-dipole method
 157 whether or not it carries the name. Nestor [17] divided a two-and-a-half-
 158 dimensional medium into cells that interact through the background Green's
 159 tensor, each carrying a contrast renormalised by its own near field, with
 160 the multiple-scattering series summed. Kanaun and Levin [18] discretise the
 161 inclusion problem with Gaussian approximating functions whose matrix ele-
 162 ments are analytic. Yang et al. [19] solve a stress-velocity form of the integral
 163 equation on a grid by conjugate gradients with fast Fourier transforms, and
 164 Touhei [20] does the same for a half space with a generalised Fourier trans-
 165 form. Jakobsen [21] and Jakobsen et al. [22] build the seismic forward and
 166 inverse problems on the transition operator of the discretised Lippmann-

167 Schwinger equation, in the anisotropic elastic case with a nine-component
 168 state vector. The elastic problem adds a structure the scalar and electro-
 169 magnetic ones lack. A cubic voxel responds to the incident field with a force
 170 (through its density contrast) and a moment (through its stiffness contrast),
 171 so the state at a voxel has nine components (displacement and strain), and
 172 the single site's self-interaction is the elastic depolarisation of a cube, the
 173 discrete counterpart of the inclusion tensor of Eshelby [23].

174 *This paper.* It extends the formulation of Nestor [17] to cubic voxels, and
 175 is ordered as the experiment is built. §2 sets out the discrete medium and
 176 §3 the continuous layer it must reproduce. §4 shows why the point coupling
 177 fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives
 178 the single-site T -matrix from moment integrals, tests it against the exact
 179 sphere, and shows that with it the discrete system is a collocation of the
 180 continuum equation. The wavefield basis comes next: §7 gives the second-
 181 order error of the uniform field in closed form, and §8 the voxel whose field
 182 carries Legendre moments to degree p , with its order $2p+2$ and its constant,
 183 for incident P and S at normal and oblique incidence. The medium basis
 184 follows in §9: a randomly stratified layer, for which a constant in each cell
 185 is exact, and a medium that varies within a cell, for which the contrast is
 186 expanded to degree r , the order becomes $\min(2p+2, 2r+2)$ and the error of
 187 the single-site T -matrix is the relative projection error of the medium. §10
 188 carries the prediction to a sphere, §11 discusses what the results imply for
 189 voxel schemes in general, and §12 closes with why an error known in advance
 190 is worth having. Every result is derived symbolically in Mathematica and
 191 checked against an independent Python implementation or an exact solution.

192 2. Setting

193 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 194 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 195 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 196 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 197 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 198 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

199 displacement followed by engineering strain. A voxel responds with a force
 200 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 201 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 202 voxel's force and moment to the state at voxel i . The Foldy-Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

203 with ψ^0 the incident state.

204 *Where the kernel comes from.* Write the medium as the background plus
 205 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 206 $\rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Split-
 207 ting $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta\lambda \delta_{bc} \delta_{de} + \Delta\mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$
 208 the contrast stiffness, and moving the contrast terms to the right leaves the
 209 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta\rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

210 a force from the density contrast, and the divergence of the contrast stress m ,
 211 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the

background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

the boundary term vanishing because m is confined to the scatterers, and the last step holding because $\mathbf{G}$ depends only on the separation. The moment therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when the moment is the contrast stress itself. With the moment tensor defined instead through its equivalent body force $-\partial_c[M_{bc}\delta(\mathbf{x}-\mathbf{y})]$, as in Aki and Richards [24], the same field reads $M_{bc}\partial G_{ab}/\partial y_c$, the source derivative, with $M = -m$. The two are one convention written two ways; what matters is that the derivative in the kernel and the sign in the contrast operator are chosen together. The nine-component kernel follows (Fig. 1): its force columns are $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are those entries and its strain rows their receiver derivatives, symmetrised in the same way and doubled on the shear rows to give engineering strain. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$ and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

Averaging, and collocation. Three operations in this paper are averages over a cell, and they are distinct. The *source-cell average* integrates the kernel over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) \, d\mathbf{y}$ is the field at $\mathbf{r}$

233 of a uniform polarisation of the source cell. Every scheme in this paper uses
 234 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 235 is imposed: at the receiving cell’s centre, with the internal field of each cell
 236 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 237 *average* imposes the equation in the mean over the receiving cell instead;
 238 with a uniform internal field that is the Galerkin scheme of degree zero,
 239 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 240 term that refers to the internal field and not to the test, and they differ only
 241 in the test function (a delta at the centre against the cell’s indicator) and
 242 in the constant of their $O((kd)^2)$ error (Table 3). The *cell-mean contrast*,
 243 finally, is an average of the material, not of the field: where the medium
 244 varies within a cell, the contrast is projected onto the cell’s basis, its mean
 245 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 246 of the coupled-dipole method, the point dipole is collocation with the point
 247 kernel, the integrated Green tensor adds the source-cell average, and the
 248 filtered coupled dipoles replace the point source by one of finite extent.

249 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 250 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 251 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 252 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 253 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

254 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 255 join the centres of the voxels in one plane (a square lattice of pitch d), and

256 m is the separation of the two planes in units of d (Fig. 2a,b). The question
 257 is whether, and how fast, this chain converges to the continuous layer as
 258 $n \rightarrow \infty$ at fixed D .

259 *Implementations.* Every result is derived symbolically in Mathematica and
 260 compared with an independent numerical implementation or an exact solu-
 261 tion. “The package” below is the Python implementation of the voxel scheme
 262 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 263 propagator, which serve both as the object under test and, where validated,
 264 as a second implementation.

265 3. The continuous layer and its thin-layer expansion

266 At normal incidence a plane force drives a one-dimensional problem per
 267 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 268 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 269 follow from continuity of displacement and traction at its faces; they agree
 270 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 271 P and S (`ContinuumLimit_Reference.wl`).

272 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 273 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 274 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer’s wavenumber. The reflection coeffi-
 275 cient expands as

$$\begin{aligned}
 R = & \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\
 & + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5),
 \end{aligned}
 \tag{7}$$

276 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 277 coefficient symbolically and the truncated series against the exact layer).
 278 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

279 to first order a thin layer is an interface across which the traction jumps by
 280 the excess inertia and the displacement by the excess compliance. At oblique
 281 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 282 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 283 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients
 284 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 285 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 286 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 287 a single plane of voxels must reproduce order by order. A plane of voxels
 288 that carry Legendre moments of their internal field to degree p reproduces
 289 Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 290 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 291 first-moment voxel all four terms shown.

292 4. The specular sums of the point Green's tensor

293 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 294 of every voxel of a plane m planes away. In that form it is a poor object to
 295 study. The Green's tensor decays slowly with distance, so the sum converges
 296 slowly, and it hides the one question that matters here: which part of it is
 297 the continuum, and which part is the discretisation. The Poisson summation

298 formula [25] answers both at once, and is the standard tool for lattice sums
 299 of wave fields [26]. For a function f on the plane and the square lattice of
 300 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

301 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 302 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3).
 303 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

304 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 305 between planes a distance z apart. Each term has a physical meaning. Trans-
 306 formed laterally, the field of a plane of sources is a superposition of plane
 307 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 308 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 309 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 310 far beyond the wavenumber of the field when the voxels are small compared
 311 with the wavelength, so it is evanescent and decays away from the plane as
 312 $e^{-|\mathbf{g}|md} = e^{-2\pi m \sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 313 the lattice sum exactly into the continuum and a discretisation correction,
 314 and the correction converges after a few orders.

315 *The local term that survives.* In the strain-from-moment block the evanescent
 316 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 317 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum

term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
planes in slab order, the static lattice sum of the point kernel is exactly
the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
on the normal shears, per unit volume) plus a cubic-symmetric remainder
(`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
reproduces the continuum only if that remainder is cancelled by the voxel's
own self term.

5. The tiling identity

Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
Each Poisson order in Eq. (10) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 4).
Two exact statements follow.

Between planes. For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

the continuum's plane-wave term times the vertical form factor. The pack-
age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
`AveragedSums.wl`).

338 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 339 is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

340 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 341 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 342 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 343 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 344 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 345 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 346 with every cell integral evaluated analytically (below).

347 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 348 same-plane block of the cell-averaged Foldy-Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

349 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 350 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 351 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 352 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 353 and the quadrature order are raised; Eq. (14) is now the package's kernel at
 354 normal incidence.

355 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 356 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 357 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 358 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,

359 which never contain the origin, and there the two-dimensional antiderivative
 360 in the face coordinates is continuous, so the face integral is a sum over the
 361 face's corners. The distributional term at $r = 0$ is carried by the divergence
 362 theorem itself. (A triple antiderivative evaluated at the cube's corners is
 363 *not* usable for repeated derivatives: its arctangent terms jump across the
 364 coordinate planes through the origin, and differentiating twice there discards
 365 exactly that term.) The self term so obtained reproduces the classical closed
 366 form of the cube's depolarisation to machine precision.

367 **6. The single site: the cube T -matrix from moment integrals**

368 Every error of a voxel scheme is made at the single site (§11), so the single
 369 site is derived here rather than imported. The classical route imports it. For
 370 an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a
 371 uniform load uniform, with a depolarisation tensor in closed form [23, 27, 28];
 372 its dynamic generalisation is known for spheres and ellipsoids [29, 30]. For
 373 a cube or a general polyhedron the static field of a uniform eigenstrain is
 374 also in closed form [31, 32, 33, 34], but the strain it produces is not uniform,
 375 so there is no exact uniform-strain inclusion to import. In scattering theory
 376 the same object appears as the self-consistent or effective-field closure of
 377 the volume integral equation [35, 16, 36]: a first-order (Born) scatterer [37]
 378 renormalised by the geometric series of its own near field, the step that turns
 379 a contrast into an effective contrast. Nestor [17] introduced the single site
 380 in this form, summing the series of the cell's self integral, with the singular
 381 near field evaluated over an infinitesimal cell. The moments below evaluate
 382 the same self integral over the finite cube, as a distribution. The elastic T -

matrix of Waterman [38] represents a scatterer in spherical multipoles, the natural basis for a sphere; a voxel in a lattice needs one compatible with the lattice, and the moments over the cube provide it. The construction below makes the approximation of a voxel explicit rather than implicit: it derives the cube's T -matrix from the moments of the elastodynamic Green's tensor over the cube, and identifies the uniform-strain closure as one truncation of an exact hierarchy.

6.1. The closed hierarchy

Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q . The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (16)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

all fields evaluated at the centre. Its coefficients are six moments of the background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (18)$$

398 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 399 reaches **N** and **Q** only, and no symmetry, parity or scaling check can detect
 400 it, so it is fixed by convention. The moments are graded by (D, W) , the
 401 number of derivatives and the degree of the weight: **G** $(0, 0)$, **K** $(0, 2)$, **N** $(1, 1)$,
 402 **M** $(2, 0)$, **P** $(2, 2)$, **Q** $(3, 1)$.

403 6.2. The moments as distributions

404 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (19)$$

405 [39, 24]. Expanding g_c in powers of r reduces every moment in Eq. (18) to a
 406 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (20)$$

407 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 408 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 409 gral depends on the shape of the region excluded around the origin. This is
 410 the source of the δ -function term of the inclusion problem, which a numerical
 411 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 412 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 413 are distributional from the start, and each is moved onto the voxel's faces by
 414 pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (21)$$

415 The surface term is evaluated on the faces, a distance h from the origin, where
 416 every derivative of r^m is smooth. The recursion lowers D and W together

417 and ends on convergent integrals. The face integrals are done in closed form
 418 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (22)$$

419 which the face route satisfies exactly and a spherical excision returns as
 420 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell
 421 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 422 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 423 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 424 Because every power of r is paired this way, the dynamic moments keep the
 425 even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones.
 426 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 427 those terms, at a real $O((kh)^2)$ cost (§6.4).

428 6.3. Parity, and the nine-component block

429 The cube is centrosymmetric. A moment that couples a field quantity of
 430 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 431 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (16) contains the first gradient alone: ∂u is
 432 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 433 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 434 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 435 by Eq. (16), and *no treatment of the second-gradient block (17) can change*
 436 *it*. What Eq. (17) contributes reaches the far field only through the $O((kh)^2)$
 437 coupling in Eq. (15). That is the order, and the origin (the variation of the
 438 field across the voxel), of the collocation error of §7, which the first-moment
 439 voxel of §8 removes.

440 With $\Delta\rho = 0$, Eq. (16) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 441 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 442 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 443 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (23)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (24)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (25)$$

444 The density channel is the displacement block of Eq. (15) with the second
 445 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 446 of $\mathbf{G}$.

447 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 448 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 449 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 450 latational channel, Eq. (23), is the sphere's exactly: its eigenvalue involves
 451 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 452 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 453 the shape. The cube's shear channels differ from each other. Their isotropic
 454 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (26)$$

455 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 456 order is therefore a pure cubic anisotropy. It splits the deviatoric channel

about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
and the diagonal shear is the more depolarised. At the background of §7
($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
0.4960.

Where Eshelby enters. Equations (23)–(25) follow from the moment integrals
and the group theory alone. No ellipsoidal result is used, and no δ term is
added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
the uniformity theorem that ansatz is exact if and only if the inclusion is
an ellipsoid. For a sphere the construction is therefore exact in the static
limit, and for a cube it is an approximation. The hierarchy (15)–(17) states
that approximation as the omitted blocks, rather than as an unquantified
shape correction. A polynomial expansion about the centre cannot represent
the edge and corner concentrations of an isolated cube's internal field at any
finite degree. A voxel in a uniform layer has no such concentrations: its
neighbours have the same properties, so its internal field is the continuum's,
which is smooth. That is why a low-degree internal field suffices in a lattice
(§§7–8) although it is only an approximation for an isolated cube.

Against the exact sphere. These statements are tested against the exact elas-
tic Mie solution for a sphere, with no Eshelby formula in between. The
concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
route of Eq. (21) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet

482 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 483 isotropic shear average, Eq. (26), is the depolarisation read from Mie's E_2
 484 to 10^{-8} . The same moment taken as a principal value, over the ball with
 485 a small sphere excluded around the origin, has zero depolarisation in every
 486 channel. The whole static response of the sphere is the δ term.

487 At finite frequency the closure uses the dynamic ball moments, which are
 488 in closed form. It departs from Mie at second order in ka and first order in
 489 the contrast, and the departure is closed-form too: the ball's moments are
 490 elementary, and Mie's low-frequency series with the contrast kept exact gives
 491 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (27)$$

492 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 493 the radiation damping, agree exactly in all three channels. So the closure's
 494 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 495 own. That is the order the hierarchy assigns to the omitted second gradi-
 496 ents (§6.3), and it has the same origin, the variation of the field across the
 497 scatterer, as the collocation error of §7. Eq. (27) is derived symbolically
 498 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 499 the numerical Mie solve and an independent implementation of the closure
 500 to 5×10^{-4} , which is the size of the next order in the contrast.

501 6.4. The collocation closure

502 The collocation closure takes the internal field of a voxel as uniform, the
 503 truncation of §6.3 for the density and stiffness channels together. The voxel's
 504 own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 505 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 506 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (28)$$

507 With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 508 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (29)$$

509 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 510 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
 511 so Eq. (29) is precisely the collocation of the continuum Lippmann–Schwinger
 512 equation with a strain piecewise constant over voxels. It does not contain
 513 $\mathbf{S}_{\text{self}}$ at all.

514 The package's cube T -matrix is this closure statically: both blocks agree
 515 with Eq. (28) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 516 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 517 through its far-field form-factor corrections. Its self integrals are exact in the
 518 static and radiation parts but omit the real even-power terms of the Green's
 519 tensor's expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 520 self term, whose series agrees with the closed-form elastodynamic tensor to
 521 10^{-14} . By Eq. (29) that error cannot enter the closure scheme.

522 7. Convergence, and the closed-form error

523 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
524 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
525 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
526 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
527 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (29) and with the
528 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
529 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
530 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
531 the exact solution and n the refinement; the orders in the legend of Fig. 5
532 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
533 with their standard errors. The two agree where a sequence is asymptotic
534 and differ where it is not, which the apparent orders show.

535 *The leading error in closed form.* In Eq. (29) the kernel is integrated exactly;
536 the only approximation is the internal field's variation across a voxel. At first
537 order in the contrast the discrete scattered field is therefore a midpoint rule
538 applied to the continuum's Born integral, with the kernel's cell integral exact.
539 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
540 it is flat. With k the background wavenumber of the wave concerned (k_P for
541 an incident P wave, k_S for an incident S wave), R the reflection coefficient
542 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (30)$$

543 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
544 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
545 and those of Eq. (33) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
546 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
547 the transmission constant grows to -0.092 .

548 *Why a uniform-field single site stops at second order.* The two constants have
549 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
550 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
551 reduce the reflection error by a factor 4.45 and *increase* the transmission
552 error by 1.7, at every n . The missing term in Eq. (30) is the first moment
553 of the internal field over each voxel, which radiates through the derivative of
554 the kernel. A fourth-order scheme must carry it: a single site with a gradient
555 channel, and a kernel whose cell integrals include the kernel's derivative.

556 8. Fourth order and beyond: Legendre moments of the voxel field

557 Let each voxel carry the mean *and* the first moment of its internal field.
558 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) =$
559 $P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w =$
560 $\sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the
561 polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the
562 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (31)$$

563 with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz\right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (32)$$

564 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
565 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$
566 and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form.
567 Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 con-
568 verges at fourth order in both reflection and transmission (`ContinuumLimit_`
`FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (32). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

569

570 the 2 m layer 6500 times more accurately than a single plane of uniform ones.
571 The collocation column is the scheme of §6 and reproduces the chain of the
572 full 9×9 system to all printed digits.

573 *Higher moments.* Carrying the Legendre moments to degree p changes noth-
574 ing in the construction: the moments and the same-cell double integrals of
575 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-

576 degree solver reproduces the chain above exactly. The second-moment voxel
577 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
578 eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment_`
579 `w1`); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

580

581 *The leading error in closed form.* Without scattering, the Galerkin solution
582 is the L^2 projection Π of the incident field onto each voxel's basis, and the
583 kernel is integrated over each voxel exactly; at first order in the contrast the
584 error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field
585 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
586 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and
587 every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] \, ds}{\int_{-h}^h e^{i(q+k)s} \, ds}, \quad (33)$$

588 with Π the centre value for collocation, the cell mean for the mean-only voxel
589 and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_`

590 `FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results
are in Table 3. The chain of Eq. (32) reproduces the forms for collocation

Table 3: The leading error of the discrete layer in closed form, Eq. (33): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

591
592 and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being
593 the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7}
594 at a contrast reduced by 10^{-6} . The constant for degree p follows from the
595 Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops
596 the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment
597 against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n/(2n+1)!!$ and
598 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (34)$$

599 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
600 mission for odd p , opposite signs for even p . The first-moment voxel's leading
601 error therefore does not depend on the direction in which the field is observed.

Eq. (34) is checked symbolically for $p = 0$ to 5 (every lower power of kd vanishes) in Mathematica and, independently, in sympy.

One plane of voxels and the thin-layer series. Refining at fixed thickness is one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$, $d = D$) the series in D of the reflection coefficient, computed exactly at the full contrast, coincides with that of the exact layer, Eq. (7), term by term through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.w1`). Each further Legendre moment adds two terms of the thin-layer series.

In three dimensions. The three-dimensional first-moment voxel carries the nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns in all, tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (32) (`ContinuumLimit_Reduction3D.w1`). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and $x/h, y/h$ are odd there. The mean and the z -moment carry weights that are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in z alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift,

626 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h / p\pi + O(k_x^2 h^2)$, so the tiling identity holds
627 only to $O(k_{\parallel} h)$.

628 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
629 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (35)$$

630 with L_a the lateral Legendre factors. The whole-space tensor decays as
631 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each
632 entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma +$
633 $\mathrm{i}\beta \text{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double
634 averaging makes the lattice sum converge without Ewald splitting, to three
635 digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20°
and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (35). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

636

637 second order and the first-moment voxel fourth order in both the reflected

638 P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$
639 and the evanescent coupling of the gradient channels do not cost an order.
640 At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (32) to all printed digits.

641 *Incident S.* Nothing in the construction is specific to P. For a normally in-
642 cident S wave the layer is the one-dimensional problem of Eq. (32) with
643 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes
644 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
645 transmission, and the full 9×9 construction with an incident SH wave re-
646 produces that one-dimensional result to all printed digits. At oblique inci-
647 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

648

649 of Eqs. (30) and (33), are those of the incident P wave.

650 The spectral construction is implemented twice. The second implementa-
651 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
652 nition of the scheme: it takes the depth dependence of the transformed kernel
653 from an independent residue construction, obtains the local weight as a nu-
654 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-

655 ble integral by quadrature and separates P from S by sampling the upgoing
 656 field at two depths. The raw reflected displacements of the two agree to $9 \times$
 657 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
 658 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
 659 scheme of any degree is also implemented twice. The second implementation
 660 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
 661 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
 662 from its own interface system, in double precision. Its errors agree with the
 663 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
 664 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 665 6.00; for one plane as the layer thins its differences from the exact layer fall
 666 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 667 Figure 5(a,b) shows the convergence, with the orders fitted to it.

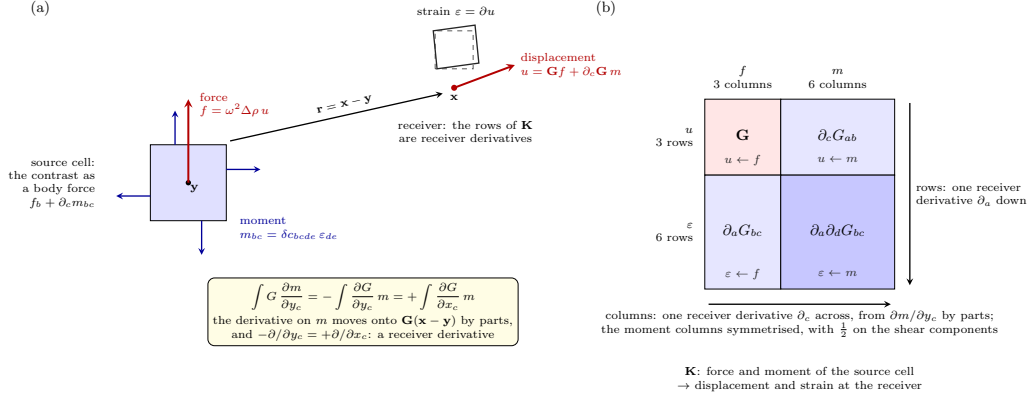


Figure 1: Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m , the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε . The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled u , three rows, and strain, six rows, and columns labelled f , three columns, and m , six columns; the blocks read $\mathbf{G}$, the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

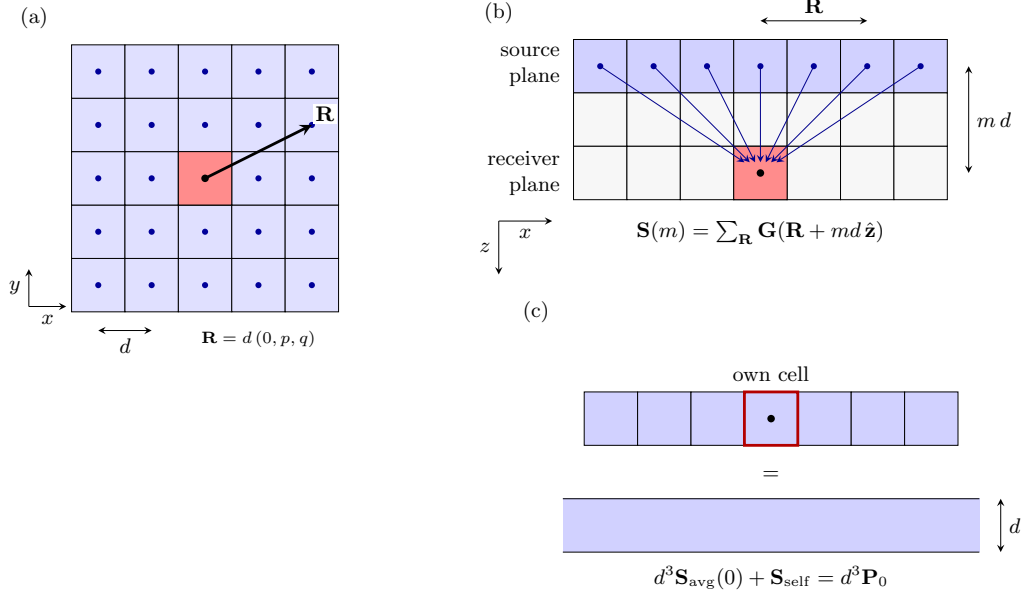


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (red) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

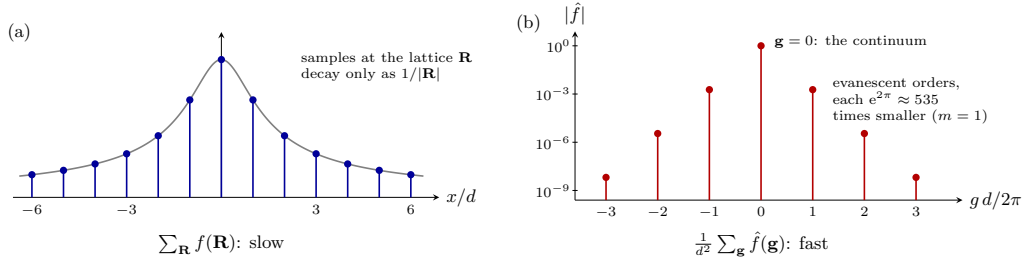


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

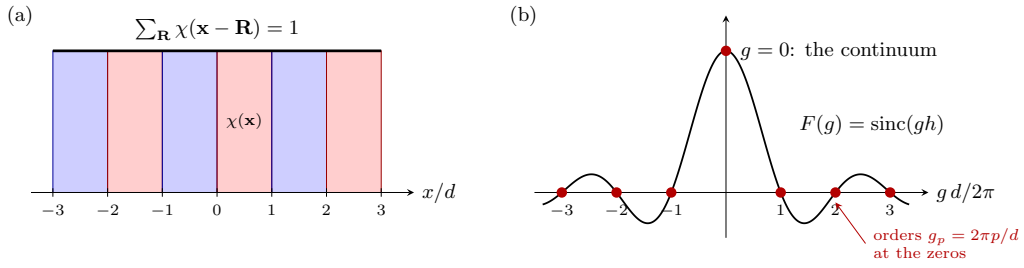


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

668 9. A heterogeneous layer

669 A heterogeneous medium is not a layer of identical cells. The simplest
670 heterogeneous case with an exact answer is a stratified model built from layers
671 of constant properties, with real interfaces between them: the form of many
672 layered crustal velocity models. The last test takes such a model. A layer
673 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
674 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
675 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
676 stratification with physical contrasts. The voxels refine the model's layering,
677 with m planes of voxels in each model layer, so no voxel straddles a model
678 interface: the natural discretisation when the model's interfaces are known.
679 At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
680 reverberates between the model interfaces. The exact answer is the product
681 of the model layers' propagators. Its same-type reflection coefficients agree
682 with those of the reflection–transmission recursion of Kennett [40], computed
683 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
684 now changing from plane to plane; nothing else changes. The errors are in
685 Table 6 and Fig. 5(c).

686 *The interfaces cost nothing.* Every order is the one measured for the uniform
687 layer. At a model interface the strain of the true field jumps. A voxel basis
688 that is discontinuous from voxel to voxel represents that jump exactly when
689 the interface lies on a voxel boundary, so the interface adds no error of its
690 own. The only requirement is that voxel boundaries coincide with the model's
691 interfaces. A voxel that straddled an interface would carry a contrast that is
692 not uniform within it, a case not examined here.

Table 6: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

693 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
 694 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
 695 layer. The first-moment voxel needs 8, one per model layer: the model's own
 696 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
 697 against 36), that is 3123 against 288 per vertical column of voxels, a factor
 698 of eleven. In three dimensions, with the lateral grid refined as the vertical
 699 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
 700 unknowns by about 2×10^4 . The same model has been solved independently
 701 in Python, with a contrast for each plane: the raw reflected displacements
 702 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
 703 at 20° , with and without the first moments. Lateral heterogeneity needs the
 704 full three-dimensional solver; it is the first extension discussed below.

705 *A medium that varies within a cell.* The stratified model is exact on its own
 706 grid: every cell is homogeneous. A smoothly varying medium is not, and
 707 sampling it as cells of constant properties is an approximation of its own.
 708 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
 709 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
 710 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
 711 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (32)
 712 becomes

$$\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (36)$$

713 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 714 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 715 the order of convergence is

$$\min(2p+2, 2r+2), \quad (37)$$

716 or $2p+2$ when degree r represents the profile exactly (Table 7; `ContinuumLimit_`
 717 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 718 caps every voxel at second order, however rich its field basis, and for the
 719 smooth profile the voxel of degree 2 with a linear contrast has the error
 720 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 721 this is what one expects: the contrast's projection error is orthogonal to
 722 the polynomials of degree r , so against the smooth product of the inci-
 723 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 724 the contrast must be represented to the degree of the field. The closed forms
 725 of the construction are tested one by one by an independent implementa-

Table 7: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (37), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
bles the scheme without linearisation, evaluating the projected contrast at
the quadrature nodes, against its own 30-digit solution of the graded layer;
its errors agree with those of Table 7 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it
measures the orders of Eq. (37) again.

Which basis sets the error. Equation (37) gives the order. The size of the
error, and which of the two bases it belongs to, appear when the scattered
field is separated by its order in the contrast: the Born term T_1 , and the
second-order term T_2 in which the field scattered once is scattered again.
Both follow from the solution at contrasts scaled by $\pm\epsilon$ and $\pm 2\epsilon$, for the

739 scheme and for the exact layer alike. With $f(z)$ the profile and Π_q the L^2
 740 projection onto the Legendre polynomials of degree at most q in each cell,
 741 define the *relative projection error* of the medium at degree q ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (38)$$

742 the fraction of the squared norm of the profile that polynomials of degree q
 743 on the cells cannot represent. It is a property of the medium and the grid
 744 alone; it involves neither the wave nor the solution. For the smooth profile
 745 and every one of the nine pairs with $p, r \leq 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (39)$$

746 to within 1.3% at $\omega = 300 \text{ rad s}^{-1}$ on 4 to 16 cells: the ratio of the two sides
 747 is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2.
 748 The error of the Born term is smaller by two to six orders of magnitude. The
 749 error of the whole is therefore that of T_2 , the projection error times $|T_2|/|T_1|$
 750 (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7} , and Table 7
 751 has 1.4×10^{-7} .

752 Equation (39) is exact in the long-wave limit, for a reason that shows what
 753 each basis is for. The strain-strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$,
 754 is local as $kD \rightarrow 0$, and the incident strain ε_0 is then uniform across the layer,
 755 so the strain scattered once is $-(\Delta M/M_P) f(z) \varepsilon_0$: it is the profile itself. A
 756 cell holds the contrast as $\Pi_r f$ and the field to degree p , so it holds the once-
 757 scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order
 758 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) dz = \|\Pi_q f\|^2, \quad (40)$$

759 the last because $\Pi_q f$ lies in the range of Π_r . The relative error is $1 -$
 760 $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$. Because the kernel is local in this limit, the second
 761 scattering takes place in the cell of the first: T_2 is the second-order term of
 762 the single-site T -matrix, and Eq. (39) is a statement about the single site.
 763 The T -matrix sums the rescattering within a cell exactly for the cell as it
 764 is represented, a contrast $\Pi_r f$ and a field of degree p ; its error is that the
 765 cell is not the medium. The coupling between cells contributes only the de-
 766 parture from the limit. Measured, the ratio of the two sides of Eq. (39)
 767 for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2} , 4.3×10^{-3} ,
 768 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24, 0.12, 0.04$ and 0.012 : as $(kD)^2$
 769 (`scripts/measure_layer_bases.py`). The pairing in Eq. (40) is verified
 770 symbolically for all nine pairs, and the projection errors are computed ex-
 771 actly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each halving
 772 of the cells.

773 Three consequences follow. The two bases must be matched: a contrast
 774 of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$
 775 have the errors of $(0, 0)$ to three digits), and a field of higher degree than
 776 the contrast improves only the Born term. The wavefield basis is there to
 777 carry what the medium imprints on the field, not only the wave: at these
 778 frequencies the wave itself varies little across a cell. And the error is known
 779 from the medium, before anything is solved: $\mathcal{E}_q$ is a property of the profile
 780 and the cells. It vanishes when the medium is piecewise constant on the cells,
 781 which is the stratified model above; there the contrast needs no more than a
 782 constant in each cell, and the degree of the field alone sets the order.

783 10. A sphere: what the layer predicts in three dimensions

784 The layer and the stratified layer are laterally uniform. A sphere is not,
785 and elastic scattering by a sphere, homogeneous [41, 42] or radially varying
786 [43], has an exact solution: it is the usual benchmark for volume-integral and
787 discrete-dipole methods. This section solves two spheres with the full three-
788 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
789 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
790 iterative solver, and a P wave is incident. For the graded sphere the voxel is
791 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
792 nine-component state, tested with the same constant, and the L^2 projection
793 of its contrast, which is its cell mean; every cell that overlaps the sphere is
794 kept, and the cells are coupled by cell-to-cell double integrals of the Green's
795 tensor, one block per orbit of the cube group [2]. For the sharp sphere
796 the voxel is the package's cube T -matrix, which differs from the collocation
797 closure only through its far-field corrections (§6.4) and has the same order,
798 each cell keeping the contrast of its centre. The far field is computed from
799 the solution for the incident plane wave itself, each voxel's displacement and
800 strain carrying the plane-wave phase at its centre; adding to that phase a
801 linear variation about the scatterer's centre, as a Taylor expansion of the
802 incident field would, counts the wave's variation twice and leaves an error of
803 order $(ka)^2$ that no refinement removes. The error is the largest difference
804 between the scattered far field and the exact one over nine scattering angles
805 from 11° to 169° , relative to the largest exact value, at a distance of 5×10^8
806 radii for the graded sphere: at 5×10^4 radii the near-field terms that the
807 asymptotic far field omits are 1.4×10^{-4} of the peak, 4% of the discretisation

808 error at 12 voxels across and most of it at 24. The sharp sphere, whose errors
 809 are a hundred times larger, is observed at 5×10^4 radii.

810 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 811 its shape with every grid. A sphere whose contrast falls smoothly to zero at
 812 its surface does not: the voxels that cross the surface carry a vanishing con-
 813 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 814 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 815 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 816 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 817 the radial equations of Takeuchi and Saito [43], integrated across the shell
 818 from the core's regular solutions and matched to outgoing waves at $r = a$.
 819 The equations are derived symbolically from the equations of motion, and
 820 the solution is checked against a sphere of homogeneous concentric shells
 821 in closed form, by unitarity and reciprocity, and by an independent imple-
 822 mentation that agrees to 2×10^{-11} (TakeuchiSaito.wl, ContinuumLimit_
 823 GradedSphere.wl, scripts/crosscheck_graded_sphere.py). Each voxel
 824 carries the mean of the contrast over its cell. Table 8 shows the error falling
 825 at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (scripts/pilot_
 826 graded_voxel_sphere.py): the apparent order between successive grids is
 827 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders
 828 of Fig. 5(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across the
 829 error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
 830 graded shell and the order is not yet asymptotic. This is the order the layer
 831 predicts for the mean-only voxel, and the first test of that prediction on a
 832 body with genuine three-dimensional structure.

833 What makes the sequence this clean is that the voxel carries a single error.
 834 The collocation voxel of the sharp sphere applied to this body, each cell at
 835 the profile's value at its centre and only the cells whose centres lie inside the
 836 sphere kept, carries two: its own internal-field error and the error of sampling
 837 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
 838 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
 839 2.0, 3.5 and 3.6 at $k_S a = 0.5$, and at $k_S a = 1$ the apparent orders are 1.7, 2.5
 840 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
 841 but its apparent order is not that of any single term and cannot be read from
 842 them. The same collocation voxel with the cell mean of the contrast, and
 843 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
 844 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
 845 second term, and with it removed the collocation and mean-only voxels differ
 846 only by their constants, as in Table 3. The cell mean, with every overlapping
 847 cell kept, represents the same body on every grid, and the order the layer
 848 predicts is then the one observed.

849 The first-moment voxel of §8, extended to three dimensions with a con-
 850 trast linear in each cell, converges at fourth order on this same sphere [2]: its
 851 apparent orders are 4.01 and 4.13 where the discretisation error is isolated
 852 at weak contrast, and at the full contrast its error with 16 voxels across is
 853 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below the mean-only voxel's on
 854 the same grid.

855 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
 856 and its voxelisation changes with every grid. Table 9 shows its error for grids
 857 of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_S a = 0.5$. The

Table 8: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

858 error tracks the staircase's volume, which is 5%, 17% and 4% too large on
 859 these grids. Rescaled by the ratio of the sphere's volume to the staircase's,
 860 it falls below 1% at $k_{\text{sa}} = 0.5$ and to at most 3% at $k_{\text{sa}} = 1.0$, but it still
 861 does not converge: what remains is the staircase's shape. A refinement study
 862 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
 863 measures the scheme.

Table 9: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

864 11. Discussion

865 Figure 5 collects the convergence of every test, with the order fitted to
866 each by least squares (`scripts/plot_convergence_orders.py`).

867 *Where the error of a voxel scheme lives.* The results separate the two ap-
868 proximations a voxel scheme makes. The *coupling* between voxels makes
869 none: averaged over the source cell it is the continuum’s cell integral ex-
870 actly, between planes by the form factor’s zeros and within a plane by the
871 tiling identity, and the self term it omits is exactly the one the single site
872 carries. Every error therefore comes from the *single site*, from the internal
873 field it assumes. For a uniform field that error is the midpoint rule’s, known
874 in closed form; for a linear field it is the next term of the same expansion.
875 The converse is the practical warning. A kernel that is not the cell integral
876 (the point Green’s tensor, or a cell average that does not match the single
877 site’s closure) leaves a local term that does not vanish as the voxels shrink
878 (§4): the evanescent lattice orders carry a static contribution of fixed size per

879 voxel volume. A scheme built that way converges to the wrong limit, and
 880 refinement cannot reveal it, because the error is scale-invariant.

881 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 882 single site from a modelling choice into a derived object with a stated error.
 883 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 884 only by the cubic split of Eq. (26). Its dynamic part departs from exact
 885 scattering by a sphere only through the real $(ka)^2$ term of Eq. (27); the
 886 radiation damping is exact. That term and the collocation error of Eq. (30)
 887 have the same origin, the variation of the field across the scatterer, and the
 888 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 889 field a voxel must represent is the smooth continuum field, not the field of
 890 an isolated cube, with its concentrations at the edges and corners.

891 *Relation to the coupled-dipole method.* The same partition appears in the
 892 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
 893 relation, the lattice dispersion relation [9], the polarisability taken from the
 894 exact static solution [10] and the integrated Green tensor [11] all adjust
 895 the single site so that the lattice reproduces the continuum. For space-
 896 filling cubes the adjustment is neither needed nor a free parameter: the tiling
 897 identity fixes the local field exactly, the self term of the collocation closure is
 898 the one that makes it cancel, and what is left is a closed-form error rather than
 899 an empirical one. That the method is a collocation of the volume integral
 900 equation was shown for the electromagnetic case by Lakhtakia [7]; Eq. (29)
 901 is the same statement for the elastic lattice with the self term evaluated over
 902 the finite cube, and it is what makes the error a midpoint-rule error. The

903 filtered coupled dipoles of Piller and Martin [12] and Yurkin et al. [13] band-
 904 limit the source to the grid; the source-cell average does the complementary
 905 thing exactly, its form factor vanishing on the reciprocal lattice, and the
 906 spectral construction of Eq. (35) is the periodic coupled-dipole method of
 907 Chaumet et al. [14] with the Galerkin double average in place of the point
 908 dipole. The second-order convergence reported for the DDA by Yurkin et al.
 909 [1] corresponds to the uniform-field voxel's, and the opposite-signed constants
 910 of Eq. (30) explain why a polarisability correction can improve one observable
 911 only at the expense of another, as the package's own form-factor corrections
 912 do in reflection and transmission. Going beyond second order requires a richer
 913 internal field, which is the point of higher-order bases for volume integral
 914 equations; the first-moment voxel is the lowest such basis, and in its spectral
 915 form the Galerkin double average makes the lattice sum converge without
 916 Ewald splitting.

917 *The tension between the two error sources.* The coupled-dipole method has
 918 two sources of error, the single site and the coupling between sites, and each
 919 has been corrected on its own: the polarisability by the Clausius–Mossotti
 920 relation, the lattice dispersion relation or the exact static limit, and the
 921 coupling by integrating or filtering the Green tensor. The two are not in-
 922 dependent. The point kernel's evanescent orders carry a static local term
 923 whose size per voxel volume does not shrink with the voxels (§4), and the
 924 single site must cancel exactly that term. A polarisability tuned for one
 925 kernel is therefore wrong for another: the lattice dispersion relation corrects
 926 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 927 tered one changes the term it must cancel. Tuning either alone moves the

928 error between the two, and refinement cannot reveal a mismatch, because
 929 the surviving term is scale-invariant. For space-filling cubes the tension is
 930 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 931 term that the closure carries (Eq. (29)), so the coupling error vanishes iden-
 932 tically and every remaining error is the single site's, the variation of the field
 933 across the cell. That error has a second tension of its own. It is a midpoint-
 934 rule error whose constants have opposite signs in reflection and transmission
 935 (Eq. (30)), so any scalar adjustment of the single site that improves one wors-
 936 ens the other; the package's far-field corrections reduce the reflection error
 937 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 938 however chosen, can remove it. What removes it is a richer internal field:
 939 the first moment lowers the error by two orders in kd in both observables at
 940 once.

941 *The cost of fourth order.* A first-moment voxel carries four times as many
 942 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 943 forms show what that buys. To reach an error ε in reflection, the uniform
 944 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 945 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 946 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 947 number of unknowns by about 5×10^4 (these are the Born-order constants;
 948 at the contrast studied the first-moment voxel's measured errors lie within
 949 5% of them, while the uniform voxel's transmission constant grows with the
 950 contrast, so the comparison is if anything conservative). The stratified model
 951 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 952 moment voxels for the same accuracy. Higher moments continue the trend

953 in one dimension: by Eq. (34) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 954 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 955 a quarter of a wavelength. A three-dimensional voxel of degree p has not
 956 been built; with the Legendre polynomials of total degree at most p it would
 957 carry 10 ($p = 2$) or 20 ($p = 3$) weights per state component, against 4 for
 958 the first-moment voxel. The gain is largest precisely where it matters: when
 959 the model varies on scales that are coarse compared with the accuracy the
 960 wavefield must have.

961 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 962 mogeneous layer and a randomly stratified one, for which the discrete system
 963 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 964 moment voxel in a full three-dimensional solver, where the gradient channels
 965 couple through the evanescent lattice orders (§8) and the tiling identity holds
 966 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 967 voxel converging at second order on a laterally heterogeneous body against
 968 an exact solution, and the companion paper [2] shows the first-moment voxel
 969 converging at fourth order there. A contrast that varies within a cell, which
 970 §9 shows is needed beyond second order in a smoothly varying medium, has
 971 a further consequence in three dimensions: its linear part is odd about the
 972 cell's centre, so it couples the single site's uniform-field and first-moment
 973 blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded
 974 voxel therefore needs at least the first-moment block; that voxel is built,
 975 and tested on the graded sphere, in the companion paper. The numerical
 976 evidence is at moderate, physically typical contrasts (up to 15% in density
 977 and moduli) and at $k_S h \leq 0.3$, the range in which the single-site T -matrix is

978 validated; the closed forms are first order in the contrast, and the departures
 979 from them at the full contrast are of second order. What is established is the
 980 *order* of convergence and its leading constant, not the behaviour at strong
 981 contrast, where the single site's own closure may limit the accuracy before
 982 the lattice coupling does. The single site's dynamic error is in closed form
 983 for a sphere only; for a cube, for which no exact solution exists, it is known
 984 through the layer (§§7–8). Two extensions follow directly. The first is the
 985 three-dimensional solver itself. The second is a stratified *background*, a lay-
 986 ered reference medium, as distinct from the stratified contrast of §9. There
 987 the lattice coupling then carries the reverberation of the layering, which is
 988 not translation invariant in depth, and whether a tiling identity holds for it
 989 is the natural next question.

990 **12. Conclusions**

991 A lattice of cubic voxels reproduces a continuous elastic medium in a pre-
 992 cise sense. Because cubes tile the plane, the source-cell-averaged coupling be-
 993 tween voxels is the continuum's cell integral exactly, and at normal incidence
 994 the lattice kernel is available in closed form with no lattice sum. The single
 995 site is derived rather than assumed. The moments of the Green's tensor over
 996 the cube, taken as distributions, give a closed hierarchy whose first-gradient
 997 truncation is the uniform-strain closure. That closure is exact for a sphere in
 998 the static limit, and it departs from exact scattering by a sphere only by a
 999 real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from
 1000 the sphere only by a cubic anisotropy of the shear response. The collocation
 1001 closure carries the same self term as the lattice kernel, which cancels, so the

1002 discrete system is a collocation of the continuum equation. It converges at
 1003 second order, with a leading error that is a midpoint-rule error of the internal
 1004 field, known in closed form for every discretisation, and opposite in sign in
 1005 reflection and transmission. A voxel that also carries the first moment of
 1006 its internal field converges at fourth order, at normal incidence (where the
 1007 three-dimensional scheme reduces exactly to one dimension) and at oblique
 1008 incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the
 1009 direction of observation. With Legendre moments to degree p it converges at
 1010 order $2p + 2$ with the closed-form constant of Eq. (34), and a single plane of
 1011 such voxels reproduces the thin-layer series term by term through $O(D^{2p+2})$.
 1012 Where the medium varies within a cell, a contrast sampled as constant caps
 1013 every voxel at second order; with the contrast projected to degree r the order
 1014 is $\min(2p+2, 2r+2)$, and the coupling needs nothing beyond the same kernel
 1015 moments to degree $p + r$. The orders survive the interfaces of a randomly
 1016 stratified model, provided voxel boundaries coincide with its interfaces, and
 1017 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel
 1018 per model layer, where the uniform-field voxel needs about forty. A sphere
 1019 whose contrast falls smoothly to zero at its surface converges at second order
 1020 against its exact solution, as the layer predicts, to 0.018% at $k_S a = 0.5$ with
 1021 twenty-four voxels across; a sharp sphere's error is set by its staircase, which
 1022 changes with the grid. The results, and the evidence for each, are collected
 1023 in Table 10.

1024 *The error is known in advance.* For every discretisation studied here the
 1025 error is not something to be discovered by refining until the answer stops
 1026 changing. Its order and its leading constant are known before anything is

Table 10: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (21)	sum rule (22) exact; M, Q equal an independent archive, 81/81
Channels (23)–(25)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (26)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (27)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (29)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (30)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (32)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (35)	order 4, P and S, 20° and 40°

1027 computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the
 1028 collocation voxel (Eq. (30)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the
 1029 voxels of degree 0, 1 and 2 (Eq. (34)), and the order $\min(2p+2, 2r+2)$
 1030 where the medium varies within a cell (Eq. (37)). That is worth more than
 1031 the order alone, for three reasons. First, the cell size and the degree of the
 1032 voxel that a required accuracy needs follow from the wavenumber and the
 1033 medium, before the computation and without a refinement study. Secondly,
 1034 it turns a refinement study from a hope into a test. A scheme of this kind can
 1035 converge smoothly to the wrong limit (§4), and smooth convergence alone
 1036 cannot reveal it; but the difference between two grids must have the known
 1037 order and the known constant, and a computation whose differences do not
 1038 is in error from some cause other than the discretisation. Thirdly, it shows
 1039 where effort pays. The error is that of representing the field and the medium
 1040 within a cell, so raising the degree of the cell does what multiplying the cells
 1041 cannot: in the stratified model one first-moment voxel per layer reaches 10^{-6}
 1042 where the uniform-field voxel needs about forty. Whether the error of a body
 1043 with three-dimensional structure can be known in advance in the same way
 1044 is the question taken up in the companion paper [2].

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 1047 the Research School of Earth Sciences of the Australian National University,
 1048 Canberra [17].

1049 **Declaration of competing interest**

1050 The author declares no known competing financial interests or personal
1051 relationships that could have appeared to influence the work reported in this
1052 paper.

1053 **Declaration of generative AI and AI-assisted technologies in the** 1054 **manuscript preparation process**

1055 During the preparation of this work, the author used Claude (Anthropic)
1056 in order to check the existing Mathematica notebooks, Python and Fortran
1057 code, to build tests to ensure their correctness, and to write the TikZ code
1058 of the diagrams. After using this tool, the author reviewed and edited the
1059 content as needed and takes full responsibility for the content of the published
1060 article.

1061 **Data availability**

1062 The Mathematica notebooks and Python scripts that derive and check
1063 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)
1064 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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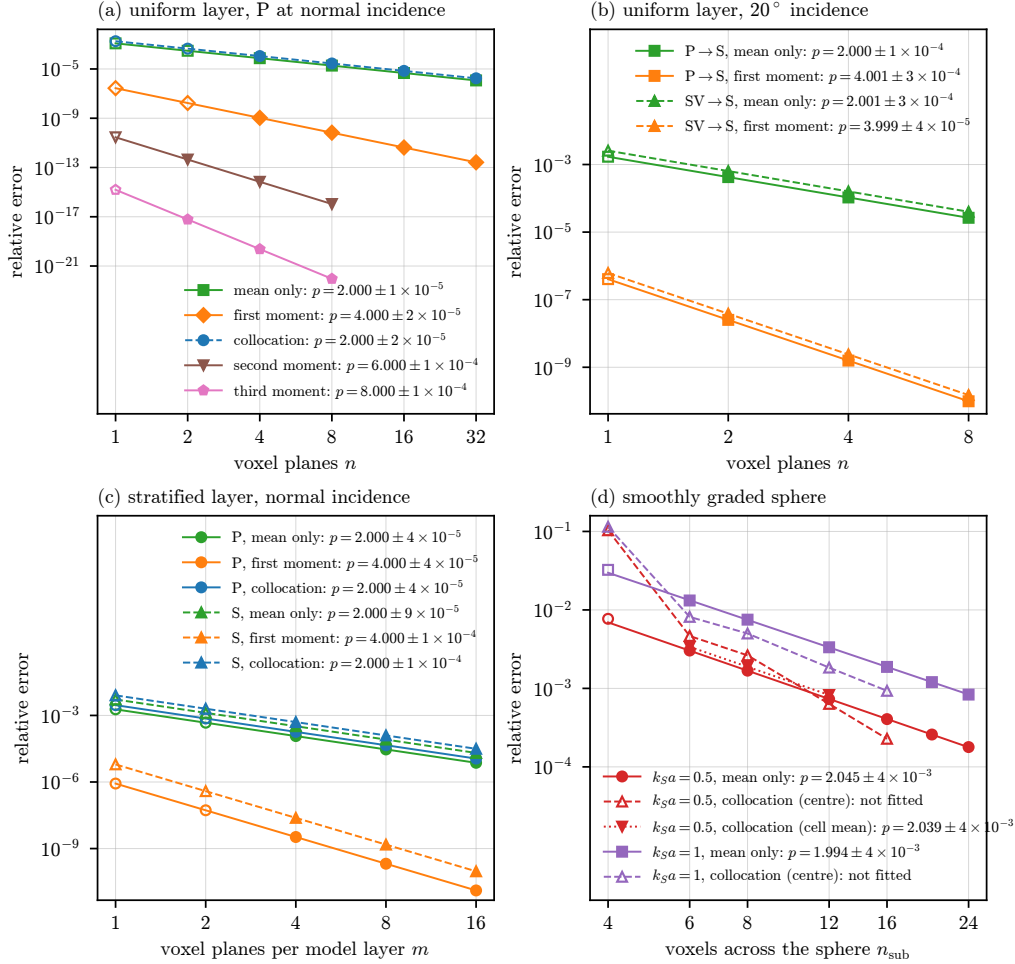


Figure 5: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. Colour marks the scheme: collocation (§6), the mean-only voxel and the voxels carrying Legendre moments to degree $p = 1, 2$ and 3 (Eqs. (32) and (35)). (a) The uniform layer, P reflection at normal incidence ($D = 2 \text{ m}$, $\omega = 300 \text{ rad s}^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor $3/2$ of Table 3; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and not fitted, since its two $O(h^2)$ errors of opposite sign leave no single order to fit; and, at $k_{\text{sa}} = 0.5$, the same collocation voxel with the